

Ricardo (Zhicong) Xie

Building Services / HVAC Intern | MVHR/ERV, CAD and Commissioning

Auckland, New Zealand | 02902729390 | zxie588@aucklanduni.ac.nz | Portfolio: <https://zxie588-alt.github.io> | GitHub: <https://github.com/zxie588-alt>

Valid NZ student visa | No sponsorship required | 25h/week during semester | Full-time 17 Nov 2026 - late Feb 2027

PROFILE

- Mechanical engineering master's student in Auckland with a New Zealand residential MVHR/ERV project covering layout, airflow schedule, duct velocity, pressure-loss estimate, equipment selection and commissioning checks.
- Interested in building services, HVAC/MEP, sustainability, facilities and consulting internships where drawings, assumptions and clear communication matter.
- Project screenshots and source links: <https://zxie588-alt.github.io>

EDUCATION

University of Auckland - Master of Engineering Studies, Mechanical Engineering | 2026 - Present

Current study and project work in building services, smart manufacturing, CFD/thermal modelling, composite structures and engineering methods.

Relevant coursework: Building Services Engineering, Thermal Modelling, HVAC System Design, Engineering Methods

Huazhong Agricultural University - Bachelor of Engineering, Mechanical Engineering / Mechatronics | 2021 - 2025

Mechanical design, CAD, electrotechnics, sensors, embedded control and mechatronic system integration.

PRACTICAL AND TEAM EXPERIENCE

- Manufacturing workshop training: prepared AutoCAD/DXF laser-cut profiles, compared drawing intent with cut parts, and worked around CNC workflow, workholding, chip formation and shop-floor safety awareness.
- Engineering practice: interpreted force-displacement test output, practised robotic arm control, and acted as a small-group lead explaining automotive seat/interior structures to peers.
- Technical visual communication: former student news-department lead with photography/event documentation experience; professional working English and native Mandarin.

PROJECT WORK

NZ Residential MVHR / ERV Ventilation Design

SolidWorks, AutoCAD/DXF, HVAC sizing, NZBC G4/H1 context, commissioning plan

- Developed a New Zealand residential MVHR/ERV ventilation design aligned with local building-code requirements.
- Modelled the MVHR unit, supply/extract/outdoor/exhaust ducts, diffusers, grilles and hangers in SolidWorks; exported DXF, STEP, STL, PNG and a component list for review.
- Sized the system for 60 L/s balanced supply/extract in normal operation and 85 L/s boost extract mode, calculating external static pressure targets of about 80 Pa and 120 Pa.
- Checked the design against NZBC G4/H1 and Healthy Homes ventilation context; developed a commissioning test plan covering CO₂, humidity, airflow and fan-power measurements.

Thermal-Fluid Simulation & Validation - Solar Chimney Case Study

ANSYS Fluent/CFX, conjugate heat transfer, natural convection

- Built a CFD validation workflow for a laboratory-scale solar chimney with coupled air, absorber and cover domains.
- Validated model credibility through mesh-sensitivity checks, global mass/heat balance review, Reynolds/Rayleigh scaling and solver-behaviour comparison before trusting velocity results.
- Documented model boundaries, limitations and recommended physical data points needed for future experimental correlation.

Orbital AI Data Centre Feasibility Study - Systems Lead

Stakeholders, requirements, ConOps, trade studies, risk, verification

- Led the systems-engineering workstream for a 5-person team study covering a 24-satellite LEO AI data-centre architecture.
- Set up the stakeholder map, requirements hierarchy, ConOps, architecture options, trade-study criteria, risk register and verification gates.
- Delivered a feasibility-report structure with trade-off matrices, risk logic and verification-gate framework across orbit/coverage, power, thermal rejection, communications, launch mass, cost and sustainability.

Manufacturing Workshop Training & DXF Preparation

Laser cutting, AutoCAD/DXF, CNC machining exposure, 3D printing workflow

- Prepared 2D AutoCAD/DXF geometry for laser cutting and engraving, then compared drawing intent with the physical cut plate.
- Observed and practised CNC machining workflow including workholding, spindle/tool motion, cutting fluid, chip formation and basic workshop safety protocols aligned with general WorkSafe NZ expectations.
- Reviewed machined gear and shaft components to connect CAD features with real manufacturing constraints such as tolerances, edge quality and tool access.

Engineering Practice and Automotive Structure Training

Mechanics testing, robotic arm control, automotive structure walkthrough

- Worked with mechanics testing equipment and interpreted force-displacement results from a physical test setup.
- Practised robotic arm control and basic automation interaction in a lab setting.
- Acted as a small-group lead during automotive structure training, explaining seat/interior structures and vehicle-system context to peers.

TECHNICAL SKILLS

- Proficient in: MVHR/ERV layout, airflow scheduling, duct velocity checks, pressure-loss estimate, fan-duty assumptions, equipment selection
- Proficient in: SolidWorks, AutoCAD/DXF, component lists, colour-coded layouts, commissioning workflow and project documentation
- Familiar with: NZBC G4/H1, Healthy Homes ventilation requirements, sensor-validation workflow, heat transfer and natural-convection CFD